

Project Title

Digital Library System

Introduction

A Digital Library System is a software application designed to manage library resources electronically. It allows librarians to maintain book records and users to search, issue, and return books efficiently. The system reduces manual work and improves the management of library resources.

Objectives

- To maintain a digital record of books.
- To add and manage library resources.
- To issue books to users.
- To return issued books.
- To display available books in the library.

Tools and Technologies

- Programming Language: C++
- IDE: Code::Blocks / Visual Studio / Dev-C++
- Data Structure: Vector
- Platform: Windows/Linux

System Features

1. Add New Books
2. Display All Books
3. Issue Books
4. Return Books
5. Exit System

Working of the System

The system stores book information such as Book ID, Title, Author Name, and Availability Status. Users can add books to the library, view available books, issue books, and return them when finished.

Algorithm

1. Start the program.
2. Display the main menu.
3. Accept user choice.
4. Perform the selected operation:
 - Add Book
 - Display Books
 - Issue Book
 - Return Book
5. Repeat until the user exits.
6. End the program.

Advantages

- Easy book management.
- Reduces paperwork.
- Fast searching and updating.
- User-friendly interface.
- Accurate record keeping.

Limitations

- Data is not permanently stored after closing the program.
- No user authentication system.
- Suitable for small-scale libraries only.

Future Enhancements

- File handling for permanent storage.
- User login and registration.
- Search functionality.
- Fine calculation for late returns.
- Database integration using MySQL.

Conclusion

The Digital Library System is an efficient solution for managing library operations digitally. It simplifies book management, reduces manual effort, and provides a structured way to maintain library records.